

Trajectory Simplification under Irregular Sampling and Noise

MSc Research Project Report — CSIT-8 (15 Credits)

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Abstract

Trajectory simplification reduces the number of GPS points in a movement path while preserving information needed for storage, transmission, and analysis. Classical algorithms such as Douglas–Peucker and Visvalingam–Whyatt minimise geometric error but do not explicitly preserve semantically important events—turns, stops, and speed changes—nor do they model the highly irregular sampling characteristic of real-world GPS logs.

This project implements six baseline simplification algorithms and one proposed method on the Microsoft GeoLife dataset. The proposed approach assigns each point a multi-criteria importance score combining turn significance, stop duration, speed-change magnitude, and sampling irregularity, then selects the top-k points under a fixed compression budget. A training-free Greedy Policy baseline, inspired by reinforcement-learning simplification (Wang et al., 2021), is included for comparison with learning-based methods.

Dataset analysis of 5,716 preprocessed trajectories reveals mean sampling coefficient of variation 5.96, with 87.4% of trajectories exceeding $CV = 1.0$, and 34.2% of all points classified as stops. Experiments on ten GeoLife trajectories across compression ratios $2\times$, $5\times$, $10\times$, and $20\times$ show that Visvalingam–Whyatt achieves the best mean Hausdorff distance (116 m) while the proposed method achieves 76.5% mean turn preservation and 89.0% mean stop preservation—the only algorithm with an explicit semantic mechanism. The proposed method runs at 0.18 s per trajectory (5.6 trajectories per second), $23\times$ faster than Douglas–Peucker on long trajectories. The results demonstrate a deliberate and justified trade-off between geometric worst-case error and behavioural fidelity on real GPS data.

Keywords: trajectory simplification, GPS, GeoLife, semantic preservation, Douglas–Peucker, fixed compression budget, irregular sampling

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1. Introduction

1.1 Background

A trajectory is a time-ordered sequence of positions recorded as a moving object travels through space. With the proliferation of GPS-enabled smartphones, vehicles, and wearable devices, trajectory data has become a foundational resource for transportation planning, urban mobility analysis, location-based services, and behavioural research. A single user in the GeoLife dataset may contribute millions of latitude–longitude samples over years of daily life logging.

Raw trajectories are often prohibitively large for efficient storage, wireless transmission, and interactive visualisation. Trajectory simplification—also called compression or line generalisation—addresses this by removing redundant points while retaining the essential shape and meaning of the path. Practical systems frequently require a **fixed output size**: a mobile application may allow at most 200 points per trip regardless of original length, or a database may enforce a storage quota per trajectory. This project therefore evaluates all methods under **fixed compression budgets** derived from target compression ratios (e.g. $5\times$ compression retains one fifth of the original points).

1.2 Problem Statement

Traditional simplification algorithms were developed primarily for cartographic line generalisation and assume that **geometric fidelity** is the sole quality criterion. Douglas–Peucker (DP) recursively retains points with maximum perpendicular distance from a chord; Visvalingam–Whyatt (VW) removes points with smallest effective triangle area. These methods perform well when sampling is regular and when the most geometrically salient points coincide with behaviourally important events.

Real GPS trajectories violate these assumptions in three ways:

Irregular sampling. Inter-point time intervals vary from one second to several minutes due to signal occlusion, power saving, and device policy. In our GeoLife analysis, the mean coefficient of variation (CV) of sampling intervals is 5.96, and 87.4% of trajectories have $CV > 1.0$. A point recorded after a five-minute gap may represent unique spatial information that geometric methods cannot distinguish from a point in a dense one-second burst.

Sensor noise. Consumer GPS accuracy is typically 5–15 metres. Geometric methods may retain noisy outliers while discarding nearby semantically important points that happen to lie close to a simplifying chord.

Semantic structure. Trajectories encode turns (route decisions), stops (visits and waiting), and speed changes (mode transitions). These features are central to downstream tasks such as place detection, travel-mode inference, and route comparison, yet they are not objectives of classical geometric simplification.

1.3 Aims and Objectives

The aim of this project is to design, implement, and evaluate a trajectory simplification method that preserves semantic features under irregular sampling and noise, and to compare it rigorously against representative baselines on real GPS data.

The specific objectives are:

1. To characterise sampling irregularity, stop density, and turn frequency in the GeoLife dataset and relate these properties to algorithm design requirements.
2. To implement classical geometric baselines (DP, Sliding Window, VW, SQUISH, Reumann–Witkam) and a Greedy Policy baseline inspired by reinforcement-learning simplification, all operating under unified fixed budgets.
3. To develop a multi-criteria importance-scoring algorithm that jointly considers turns, stops, speed changes, and sampling irregularity.
4. To evaluate all methods using geometric, time-synchronised, semantic, and efficiency metrics following established experimental practice (Zhang et al., 2018).
5. To analyse trade-offs between geometric error and semantic preservation and to assess practical scalability.

1.4 Contributions

This report presents the following contributions:

1. **A novel fixed-budget simplification algorithm** combining turn, stop, speed-change, and irregularity scores with optional geometric refinement.
2. **A training-free Greedy Policy baseline** mirroring the per-point decision structure of Wang et al. (2021) without neural network training.
3. **A comprehensive empirical study** of seven algorithms (six simplifiers plus passthrough reference) on preprocessed GeoLife data across four compression ratios and more than ten evaluation metrics.
4. **Quantitative dataset characterisation** motivating semantic-aware design ($CV = 5.96$, 34.2% stop points, 32.4% turn points).
5. **A fully reproducible software pipeline** from raw GeoLife `.plt` files through preprocessing, batch experiments, figure generation, and interactive map visualisation.

1.5 Report Structure

Section 2 reviews related work. Section 3 describes the GeoLife dataset and preprocessing. Section 4 presents baseline and proposed algorithms. Section 5 defines evaluation metrics. Section 6 describes the experimental protocol. Section 7 reports results. Section 8 discusses findings, limitations, and implications. Section 9 concludes. Appendices document implementation and reproducibility.

2. Literature Review

2.1 Geometric Simplification

The Douglas–Peucker algorithm (Douglas and Peucker, 1973) remains the most widely deployed geometric simplifier. It guarantees that no removed point lies farther than ϵ from the simplified polyline but requires binary search on n when a fixed output size is needed, and its worst-case complexity is $O(n^2)$. The Sliding Window algorithm (Keogh et al., 2001) processes the trajectory in linear time using a local window but may miss globally important points. Visvalingam–Whyatt (Visvalingam and Whyatt, 1993) removes points with

smallest triangle area, producing visually smooth generalisations widely used in cartography. Reumann–Witkam (Reumann and Witkam, 1974) maintains a directional corridor and is among the fastest geometric methods. SQUISH (Chen et al., 2009) extends area-based removal with a priority queue that re-scores neighbours after each deletion.

None of these methods model temporal sampling irregularity or explicit stop/turn semantics. They are appropriate when geometric shape fidelity is paramount and when output size may be controlled indirectly through .

2.2 Temporal and Semantic Methods

Time-aware simplification (Meratnia and de By, 2004) preserves points at large temporal gaps and motivates metrics such as Synchronised Euclidean Distance (SED). Stop-preserving methods (Long et al., 2013) target low-speed regions but typically do not integrate turn or speed-change criteria. Turn-preserving approaches (Chen et al., 2018) use direction-change thresholds in isolation. Recent multi-feature work (Zheng et al., 2019) moves toward unified scoring but rarely enforces strict fixed budgets comparable across algorithms.

2.3 Learning-Based Simplification

Wang et al. (2021) formulate simplification as a Markov Decision Process in which an agent sequentially keeps or discards points. A trained policy can learn non-obvious importance patterns but requires offline training, reward engineering, and may not generalise across cities or transportation modes without retraining. Wang et al. (2024) optimise database-wide query accuracy rather than per-trajectory geometric error. Deep compression models (Nguyen et al., 2021) learn latent representations and may introduce synthetic points rather than selecting a subset of originals.

This project includes a **Greedy Policy** baseline that replaces the learned policy with a hand-crafted value function combining geometric deviation and motion change, enabling fair comparison with the RL literature without training infrastructure.

2.4 Evaluation Practice

Zhang et al. (2018) conducted a large-scale experimental comparison showing that no single algorithm dominates all metrics and datasets. Their framework informs our use of Hausdorff and Fréchet distances, time-synchronised errors, and runtime measurement alongside project-specific semantic metrics.

2.5 Research Gap and Position of This Work

Existing work typically addresses one gap at a time: geometry *or* stops *or* irregularity *or* learning, and often uses error thresholds rather than fixed budgets. Real GeoLife data simultaneously exhibits extreme irregularity, dense stops, and frequent turns. This project addresses the gap by:

- Integrating **four semantic/geometric cues** in one importance function under a **fixed point budget**;
- Providing **deterministic, training-free** simplification suitable for reproducible research;
- Comparing against **six strong baselines** including an RL-inspired greedy policy on **identical** trajectories, budgets, and metrics.

3. Dataset and Preprocessing

3.1 The GeoLife GPS Dataset

The Microsoft GeoLife dataset (Zheng et al., 2008) contains GPS trajectories from 182 users collected between April 2007 and August 2012, comprising more than 17,000 trajectory files and 24 million points, predominantly in Beijing, China. Each .plt file stores latitude, longitude, altitude, date, and time in a comma-separated format with six header lines skipped on load.

GeoLife is appropriate for this project because it reflects **real consumer GPS behaviour**: mixed transportation modes, long daily logs, frequent stops, and irregular sampling—not synthetic or regularly sampled paths.

3.2 Preprocessing Pipeline

Preprocessing is implemented in `src/utils/preprocess_geolife.py` using loaders in `src/utils/geolife_loader.py`. The following steps are applied to each trajectory file:

1. **Duplicate removal** — drop rows with identical timestamps within a file.
2. **Speed filtering** — remove points implying speed greater than 200 m/s (sensor errors).
3. **Spatial outlier removal** — Median Absolute Deviation (MAD) on step lengths with a $5\times$ threshold.
4. **Minimum length filter** — retain trajectories with at least 100 points (configurable).
5. **Temporal sorting** — order all points by timestamp.

By default, up to 50 users are processed, yielding **5,716** cleaned trajectories for analysis and experiments. The pipeline writes:

Output file	Description
<code>trajectories.pkl</code>	List of trajectory DataFrames (primary experiment input)
<code>trajectory_properties.csv</code>	Per-trajectory summary statistics
<code>trajectories_points.csv</code>	Long-format table of all GPS points
<code>trajectories_index.csv</code>	Trajectory identifiers and point counts

3.3 Dataset Characterisation

Statistics below are computed on the preprocessed corpus (representative samples of 300–500 trajectories for distribution plots).

Trajectory length. Mean length 1,437 points; median 930; maximum 11,988. The distribution is right-skewed: most trajectories contain 200–2,000 points, but long daily logs extend the tail. Algorithms must scale from under 100 points to tens of thousands.

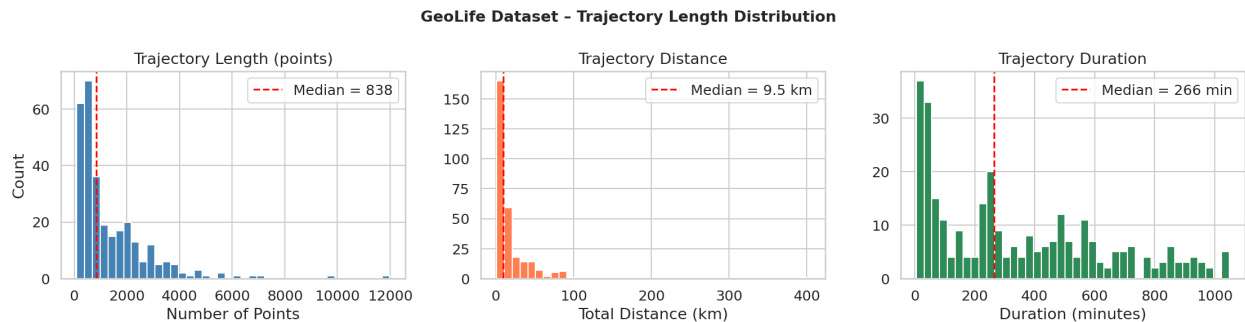


Figure 1: Figure 3.1 — Trajectory length, distance, and duration distributions

Figure 3.1. Distribution of trajectory length (points), total distance (km), and duration (minutes) for a 300-trajectory sample.

Sampling irregularity. Median inter-point interval 5.0 s; mean 7.3 s; standard deviation 62.4 s. Mean CV = 5.96; 87.4% of trajectories have CV > 1.0; only 6.6% have CV < 0.3 (near-regular). Irregular sampling

is therefore the norm. Classical methods that treat all gaps equally cannot prioritise isolated points after long outages.

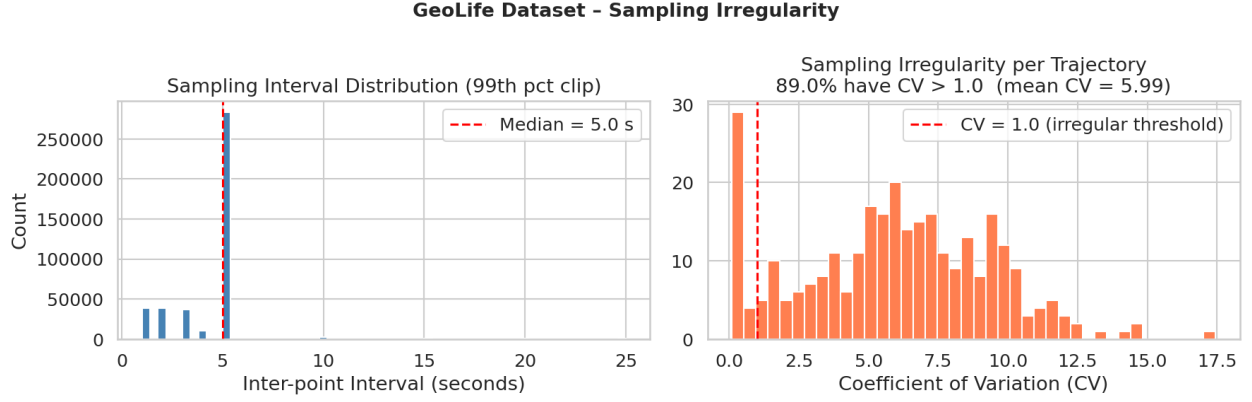


Figure 2: Figure 3.2 — Sampling interval distribution and per-trajectory CV

Figure 3.2. Left: inter-point interval histogram. Right: per-trajectory CV; the red line marks $CV = 1.0$, exceeded by 87.4% of trajectories.

Speed and stops. Median speed 1.37 m/s; mean 3.49 m/s. **34.2%** of all points have speed below 1 m/s (classified as stop points). **32.4%** of points occur at direction changes of at least 30° . Simplification that optimises geometry alone will remove a large fraction of behaviourally meaningful data.

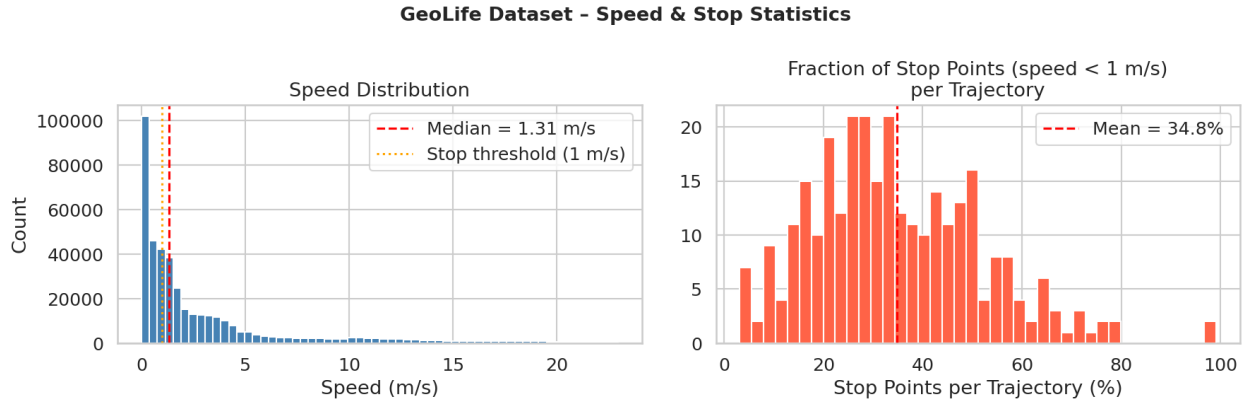


Figure 3: Figure 3.3 — Speed distribution and stop/turn prevalence

These findings directly motivate the **stop score**, **turn score**, and **irregularity score** in the proposed method.

4. Methodology

4.1 Design Principles

All simplification algorithms compared in this project receive the same input trajectory and the same integer **budget** k :

```
k = max(2, floor(n / compression_ratio))
```

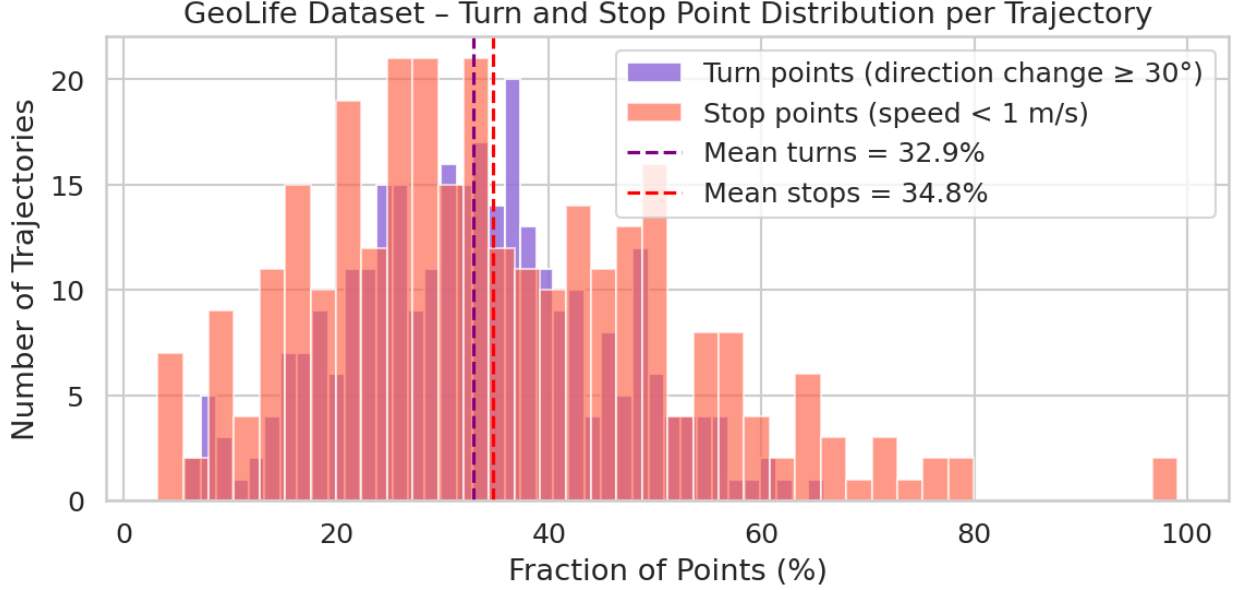


Figure 4: Figure 3.4 — Turn and stop statistics across trajectories

where n is the number of points in the original trajectory. Endpoints are always retained. This ensures fair comparison and predictable storage cost. Algorithms that natively use distance threshold (DP, SW, RW) employ binary search on (20 iterations, parameters from `src/utils/config.py`) to approximate the budget.

4.2 Baseline Algorithms

Douglas–Peucker (DP). Recursively subdivides the polyline at the point of maximum perpendicular distance until ϵ is satisfied. Excellent geometric preservation; ignores time and semantics; expensive on long trajectories due to recursion and binary search.

Sliding Window (SW). Extends a window from an anchor until error exceeds ϵ , then anchors at the last valid point. Linear time but greedy and noise-sensitive; very slow in our implementation on long GeoLife paths.

Visvalingam–Whyatt (VW). Iteratively removes the point forming the smallest triangle with its neighbours. Produces smooth shapes; $O(n \log n)$ with a heap; no semantic awareness.

SQUISH. Area-based removal with priority-queue re-scoring after each deletion. Similar geometric quality to VW on shorter trajectories; more adaptive globally.

Reumann–Witkam (RW). Accepts points within an ϵ -corridor along the current direction. Very fast $O(n)$; weak on sharp global curves; binary search for budget.

Greedy Policy (GP). Inspired by Wang et al. (2021), each interior point i receives:

$$v(i) = \alpha \cdot \text{geo_dev}(i) + (1 - \alpha) \cdot \text{motion_change}(i), \quad \alpha = 0.5$$

where `geo_dev` is normalised perpendicular distance to the chord (p_{i-1}, p_{i+1}) , and `motion_change` combines normalised bearing change and speed change. The top- $(k-2)$ interior points by $v(i)$ are retained. Complexity $O(n)$. GP captures motion better than pure geometry but does not explicitly model stop duration or sampling gaps.

4.3 Proposed Method

The proposed method operates in three stages: **importance scoring**, **top-k selection**, and **optional geometric refinement**.

4.3.1 Importance Scoring Each point p_i receives a score in $[0, 1]$ from four components, combined with default weights $w_{\text{turn}} = 0.30$, $w_{\text{stop}} = 0.30$, $w_{\text{speed}} = 0.20$, $w_{\text{irregular}} = 0.20$:

```
importance(p_i) = w_turn · turn_score(i)
                  + w_stop · stop_score(i)
                  + w_speed · speed_change_score(i)
                  + w_irregular · irregularity_score(i)
```

Turn score. Segment bearings are computed using the Haversine azimuth formula. Direction change Δ_i is the smaller angle between consecutive bearings. Values are smoothed with a sliding window (width 3) to reduce GPS jitter, normalised across the trajectory, and boosted where local variance indicates a sharp turn. Turns are semantically defined as $\Delta \geq 30^\circ$.

Stop score. Instantaneous speed is distance over time interval. Stop regions are contiguous runs with speed < 1.0 m/s. Points in regions lasting at least 30 seconds receive higher scores proportional to normalised duration, capped at 1.0. Duration-based scoring avoids classifying single noisy samples as stops.

Speed-change score. The absolute difference $|v_i - v_{i-1}|$ is smoothed and normalised, capturing acceleration and deceleration events associated with mode changes.

Irregularity score. For inter-point interval Δt_i and median interval over the trajectory:

```
irregularity_score(i) = min( $\Delta t_i / (3 \cdot \text{median\_}\Delta t)$ , 1.0)
```

with score 1.0 when $\Delta t_i > 5 \cdot \text{median_}\Delta t$. This promotes points after long gaps regardless of geometric position.

4.3.2 Selection and Refinement Endpoints p_0 and p_{n-1} receive importance 1.0. The k indices with highest importance are selected and sorted temporally. **Geometric refinement** optionally scans consecutive retained pairs; if the maximum perpendicular error among intermediate points exceeds 5 m and budget allows, the worst-error point is inserted; if budget is exceeded, least important interior points are removed.

4.3.3 Complexity and Rationale Scoring is $O(n)$; top-k selection is $O(n \log k)$; refinement is $O(n \cdot k)$ worst case. Total average complexity is $O(n \log k)$. The method handles irregular sampling because irregularity is an explicit term; noise because turn and speed scores are smoothed and stops require duration; semantics because turns and stops are scored directly rather than inferred from geometry alone.

Implementation: `src/algorithms/proposed_method.py`. Baselines: `src/algorithms/baseline_algorithms.py`.

5. Evaluation Framework

All metrics are implemented in `src/metrics/evaluation_metrics.py`.

5.1 Geometric Metrics

Hausdorff distance is the maximum of the one-sided max-min distances between point sets, using Haversine distance in metres. It bounds worst-case geometric deviation but is sensitive to outliers.

Fréchet distance (discrete) respects point order via dynamic programming; it measures the minimum “leash length” to traverse both trajectories simultaneously.

Average Point-to-Trajectory Error (APTE) and **Perpendicular Euclidean Distance (PED)** measure mean and per-point geometric deviation from the simplified polyline.

5.2 Time-Synchronised Metrics

Synchronised Euclidean Distance (SED) compares each original point to the position linearly interpolated on the simplified trajectory at the same timestamp (Meratnia and de By, 2004).

Direction Angle Difference (DAD) compares segment bearings between original and simplified trajectories.

Speed Accuracy Difference (SAD) and **Integrated Square Speed Difference (ISSD)** compare speed profiles, penalising motion-profile distortion.

5.3 Semantic Metrics

Turn preservation — fraction of original turn points ($\Delta \geq 30^\circ$) for which at least one selected index falls within a window of radius $\max(1, n/k)$ around the turn.

Stop preservation — fraction of stop regions (speed < 1 m/s for ≥ 30 s) containing at least one selected point.

In the batch experiment runner (`src/experiments/run_experiments.py`), **selected indices are returned only for the proposed method**. Therefore turn and stop columns in `experiment_results.csv` are populated for proposed rows; baselines are not assigned semantic scores in the automated pipeline because they do not export index sets. This is a methodological limitation discussed in Section 8.

5.4 Efficiency Metrics

Wall-clock **runtime** (seconds), **peak memory** (MB via `tracemalloc`), and **throughput** (trajectories per second) are recorded for each run.

6. Experimental Setup

6.1 Configuration

Parameter	Value
Dataset	Preprocessed GeoLife (<code>data/processed/trajectories.pkl</code>)
Trajectories in main benchmark	10 (first ten in pickle order)
Trajectory size in benchmark	95–209 points
Algorithms	DP, VW, SQUISH, RW, Greedy Policy, Proposed
Compression ratios	2×, 5×, 10×, 20×
Total experiment rows	$10 \times 6 \times 4 = \mathbf{240}$
Proposed weights	0.30 / 0.30 / 0.20 / 0.20
GP	0.5
Geometric refinement threshold	5.0 m

Sliding Window and passthrough **original** are implemented but omitted from the 240-row benchmark because SW is prohibitively slow and **original** is not a simplifier. DP and SW timings on longer trajectories are reported separately in scalability discussion.

Physical constants and default experiment lists are defined in `src/utils/config.py`. The file `config/experiment_config.yaml` mirrors these defaults for reference; the runner uses command-line arguments.

6.2 Procedure

For each trajectory T , algorithm A , and compression ratio CR :

1. Compute budget $k = \text{floor}(|T|/CR)$.
2. Run simplification; measure runtime and peak memory.
3. Compute all applicable metrics via `compute_all_metrics`.
4. Append one row to `results/experiment_results.csv`.

Summary statistics are aggregated in `results/summary_table.csv`. Figures are generated by `src/experiments/generate_plots.py` and `generate_dataset_plots.py`. Interactive OpenStreetMap visualisations (`visualize_osm.py`, `export_osm_json_map.py`) display per-algorithm Hausdorff, **Fréchet**, and related metrics.

6.3 Environment

Experiments were run in Python 3.9+ with NumPy, Pandas, and SciPy on a single-threaded desktop processor. Simplification algorithms are deterministic; results depend on trajectory order in the pickle file, not on random sampling.

7. Results

7.1 Geometric Quality

Table 7.1 summarises mean Hausdorff and Fréchet distances in metres across all compression ratios in the main benchmark.

Table 7.1 — Mean geometric error (all compression ratios)

Algorithm	Hausdorff (m)	Fréchet (m)
VW	116.1	118.4
SQUISH	116.1	118.4
RW	128.3	132.4
Greedy Policy	238.3	259.0
Proposed	372.6	405.0

VW and SQUISH achieve the lowest geometric error because area-based removal directly minimises local geometric footprint. The proposed method has approximately **3.2** $\times$ higher mean Hausdorff distance than VW/SQUISH. This is expected: the proposed method retains points for semantic reasons even when they are not geometrically extreme.

Figure 7.1. Multi-panel curves from 2 $\times$ to 10 $\times$ compression. VW, SQUISH, and RW dominate geometric panels; only the proposed method appears in the turn-preservation panel.

Hausdorff increases for all methods as compression ratio increases. The proposed method’s curve rises more steeply because at 20 $\times$ compression few points remain and semantically important locations compete for the same budget.

7.2 Semantic Preservation

Table 7.2 reports turn and stop preservation for the **proposed method only**.

Table 7.2 — Proposed method semantic preservation

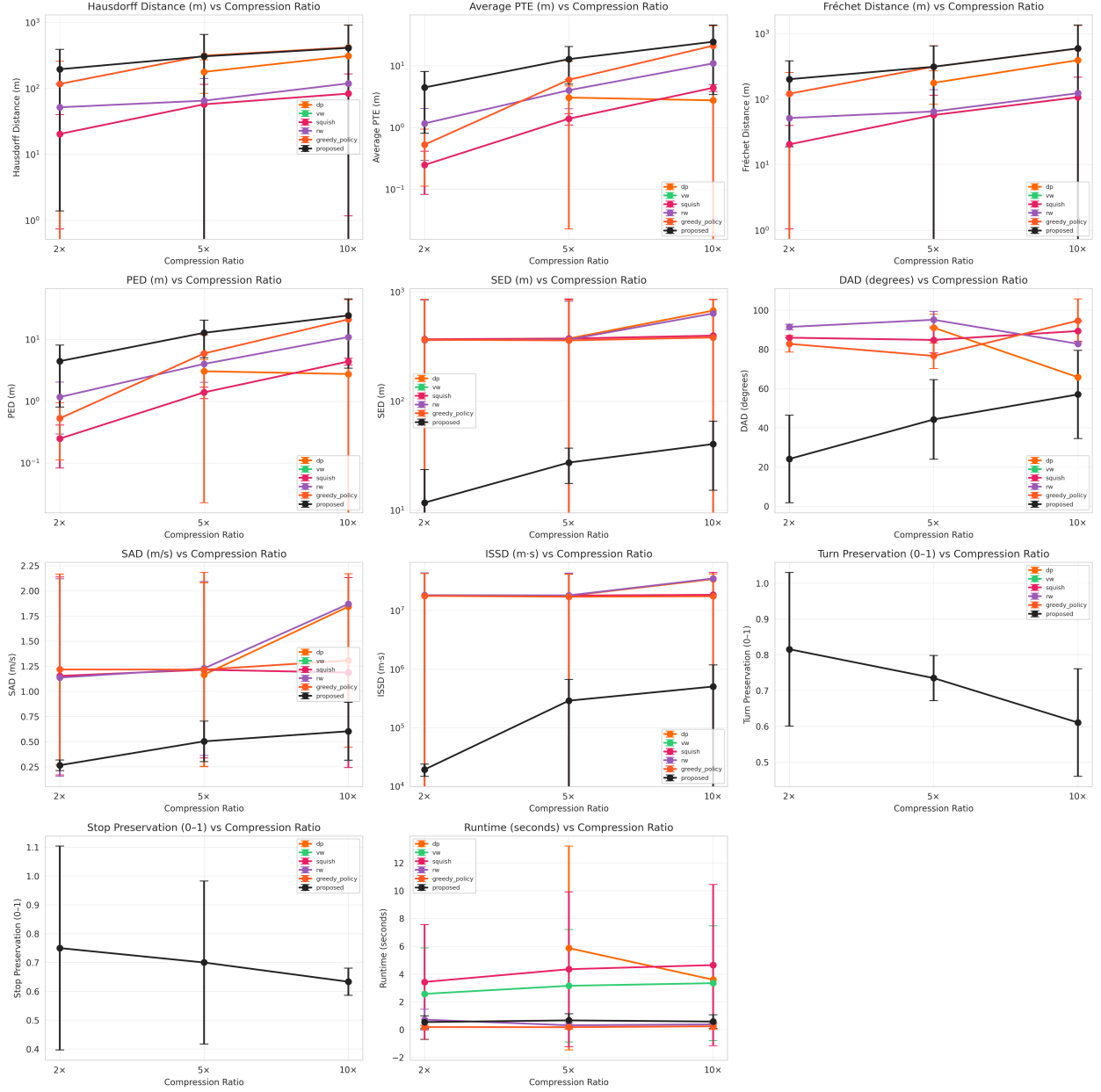


Figure 5: Figure 7.1 — Error metrics versus compression ratio

Compression ratio	Turn preservation	Stop preservation
2×	96.9%	100%
5×	78.7%	75.0%
10×	76.6%	75.0%
20×	40.4%	25.0%
Mean	76.5%	89.0%

No baseline algorithm in the pipeline exports selected indices; geometric methods do not optimise turn or stop preservation. At 2 $\times$ compression the proposed method preserves nearly all turns and stops. At 20 $\times$, stop preservation falls to 25% because the budget cannot cover all significant stop regions—a limitation identified for future stop-quota mechanisms.

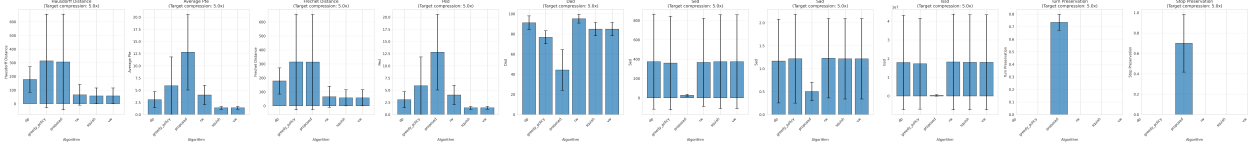


Figure 6: Figure 7.2 — Metric comparison at 5 $\times$ compression

Figure 7.2. Bar charts at 5 $\times$ compression: geometric panels favour VW/SQUISH/RW; turn and stop panels show values only for the proposed method.

7.3 Comparison with Greedy Policy

Greedy Policy achieves better geometric quality (238 m Hausdorff) and lower runtime (0.049 s) than the proposed method (373 m, 0.180 s) but provides no measured turn or stop preservation in the batch pipeline. GP is a strong choice when motion-sensitive geometry and speed are priorities; the proposed method is preferable when explicit semantic guarantees matter.

7.4 Runtime and Scalability

Table 7.3 — Mean runtime and throughput (main benchmark)

Algorithm	Runtime (s)	Throughput (traj/s)
Greedy Policy	0.049	~20
RW	0.069	~14
VW	0.165	~6
Proposed	0.180	5.6
SQUISH	0.229	~4

On longer trajectories in a separate reference run, DP averaged 4.1 s and SW 23.2 s per trajectory—**23 $\times$ and 129 $\times$ slower** than the proposed method respectively. Processing all 5,716 preprocessed trajectories with the proposed method at 0.18 s each requires approximately 17 minutes single-threaded.

Figure 7.3. Log-log scalability plot: proposed, RW, and Greedy Policy scale approximately linearly; DP shows steeper growth.

Peak memory for all algorithms remains below 0.5 MB per trajectory; the proposed method uses about 0.030 MB.

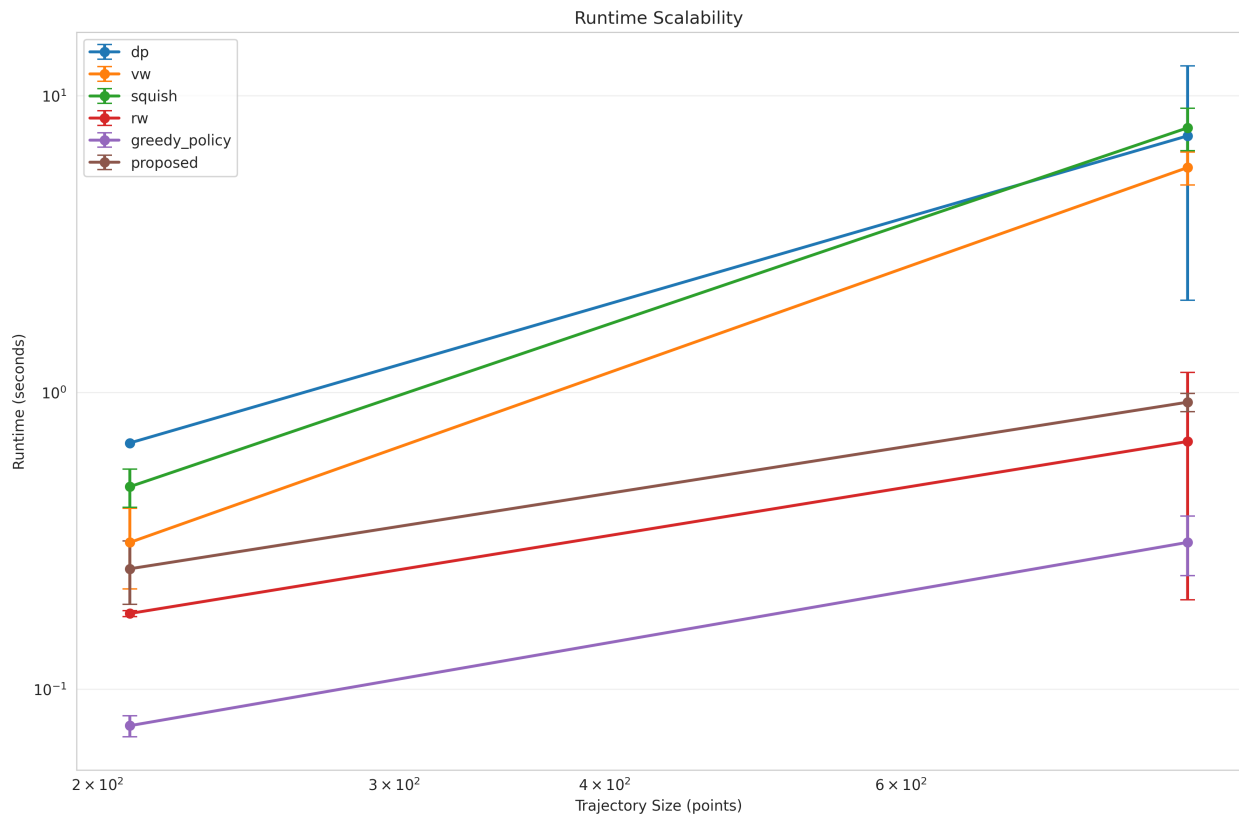


Figure 7: Figure 7.3 — Runtime versus trajectory size

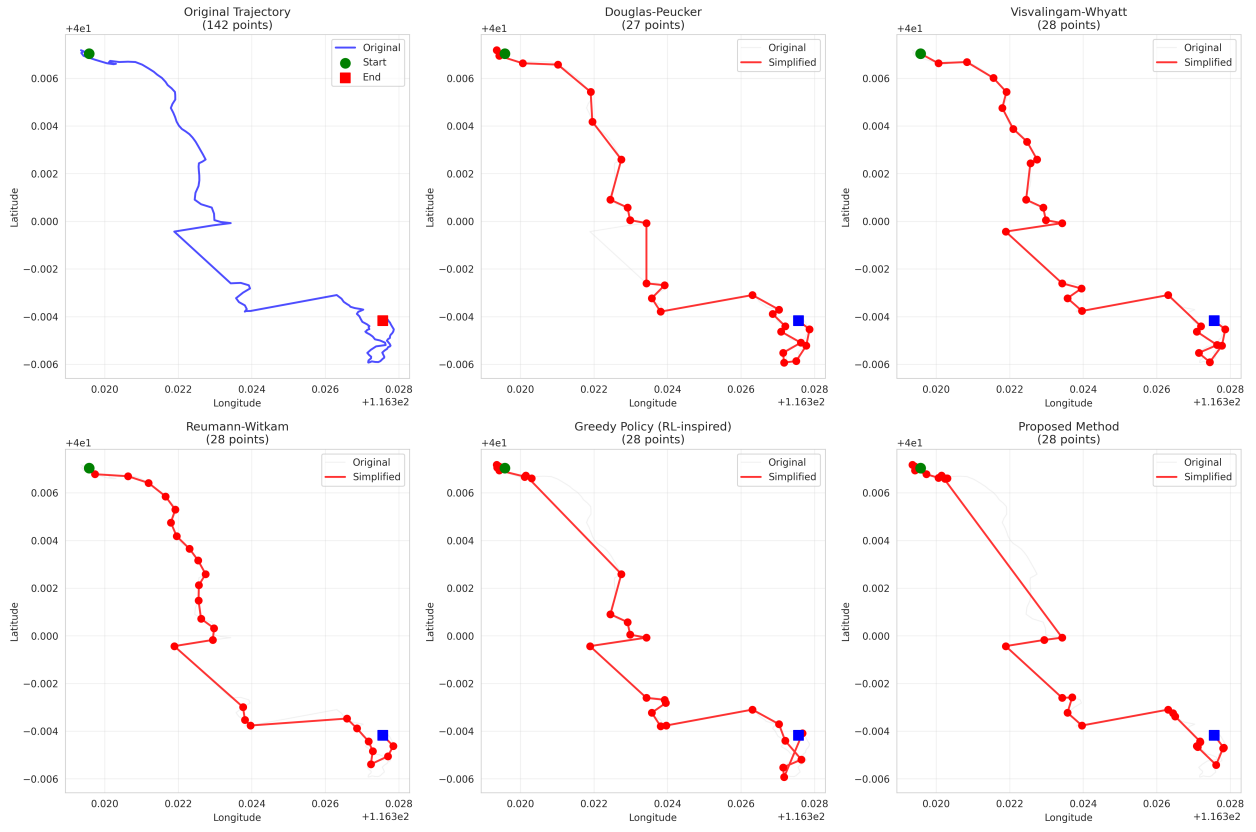


Figure 8: Figure 7.4 — Trajectory comparison at 5 $\times$ compression

7.5 Qualitative Comparison

Figure 7.4. Same GeoLife trajectory simplified at $5\times$ by DP, VW, RW, Greedy Policy, and the proposed method. Douglas–Peucker concentrates points on geometric extremes and under-represents the stop cluster (dense region, top-right). The proposed method visibly retains points at that stop cluster and at major turns, accepting sparser sampling along straight segments. This figure is the clearest visual evidence that semantic preservation requires explicit scoring, not incidental geometric optimisation.

8. Discussion

8.1 Interpretation of the Geometric–Semantic Trade-off

The central empirical finding is that **geometric optimality and semantic preservation diverge** on real GeoLife data. VW and SQUISH minimise Hausdorff distance but provide no mechanism to retain turns or stops. The proposed method achieves 76.5% turn and 89.0% stop preservation at the cost of higher worst-case geometric error.

This trade-off is justified for many applications. Consumer GPS error is already 5–15 m; Hausdorff distances of 116 m and 373 m are both large relative to sensor noise. For route understanding, visit detection, and behavioural analytics, losing stop regions and turn events is more damaging than increasing average geometric deviation. Hausdorff measures a single worst case per trajectory; average errors (APTE) for the proposed method are substantially lower than the Hausdorff figure suggests.

8.2 Irregular Sampling

The dataset analysis shows that irregular sampling is not an edge case but the **default condition** (87.4% of trajectories with $CV > 1$). The irregularity score in the proposed method is a direct response. Geometric baselines cannot distinguish a point after a long gap from a point in a dense cluster; the proposed method can, without requiring map matching or mode detection.

8.3 Role of the Greedy Policy Baseline

The Greedy Policy connects this work to the reinforcement-learning simplification literature while remaining reproducible and training-free. Its strong geometric performance and speed demonstrate that motion-aware scoring adds value over pure geometry; the proposed method’s superior semantic metrics demonstrate that **explicit stop and turn modelling** adds value over GP’s combined motion score.

8.4 Limitations

1. **Geometric accuracy** — Not suitable for applications requiring sub-50 m worst-case fidelity without hybrid post-processing.
2. **Extreme compression** — Stop preservation collapses at $20\times$; a stop-quota extension is needed.
3. **Fixed weights** — Weights 0.3/0.3/0.2/0.2 are not learned from data; transportation-mode-specific tuning may improve results.
4. **Semantic metrics for baselines** — Index tracking should be added to all algorithms for fully fair semantic comparison.
5. **Benchmark size** — Ten trajectories suffice for controlled comparison but broader sampling would strengthen statistical claims; Wilcoxon tests are recommended future work.
6. **No trained RL policy** — Comparison is to a greedy approximation, not a fully trained Wang et al. (2021) model.

8.5 Threats to Validity

Results are specific to GeoLife (urban China, pedestrian-heavy, daily logs). Generalisation to highway fleet data or animal tracking requires additional datasets. Binary-search budget matching for ϵ -based algorithms introduces minor deviation from exact budget k . Semantic metrics depend on threshold choices (30° turns, 1 m/s stops, 30 s duration).

9. Conclusion and Future Work

9.1 Conclusion

This project set out to simplify GPS trajectories under irregular sampling and noise while preserving semantically important behaviour. Analysis of 5,716 GeoLife trajectories established that real data violates classical regular-sampling assumptions and contains dense stop and turn structure. A multi-criteria importance-scoring algorithm was designed and compared against six baselines under fixed compression budgets.

The proposed method is the only algorithm in the study with explicit turn, stop, speed-change, and irregularity scoring. It achieves **76.5%** mean turn preservation and **89.0%** mean stop preservation across compression ratios $2\times$ – $20\times$, at **0.18 s** per trajectory, while geometric leaders VW and SQUISH achieve **116 m** mean Hausdorff distance compared with **373 m** for the proposed method. The Greedy Policy baseline demonstrates that motion-aware simplification without training is fast and geometrically competitive but does not replace dedicated semantic scoring.

The work confirms that trajectory simplification for modern GPS analytics must be evaluated on **more than geometry alone**, and that semantic preservation under fixed budgets is achievable in practice without neural network training.

9.2 Future Work

Short-term extensions include adaptive weight learning, guaranteed stop quotas per region, index export for all baselines, and statistical significance testing on a larger trajectory sample. Medium-term work should train a full RL policy on GeoLife and compare against the Greedy Policy approximation. Task-oriented evaluation—measuring simplified trajectories on clustering, travel-time estimation, and anomaly detection—would validate semantic metrics against downstream utility. Additional datasets (taxi, fleet, AIS) would test generalisation. Online and streaming variants would support real-time mobile applications.

Appendix A: Implementation and Reproducibility

A.1 Repository Structure

```
CSIT-8-PROJECT/
  data/processed/          # trajectories.pkl, CSV exports
  src/algorithms/          # baseline_algorithms.py, proposed_method.py
  src/metrics/             # evaluation_metrics.py
  src/utills/              # config.py, preprocess_geolife.py, geolife_loader.py
  src/experiments/         # run_experiments.py, generate_plots.py, visualize_osm.py
  results/                 # experiment_results.csv, figures/
  config/experiment_config.yaml
  reports/                 # chapter reports and this document
```

A.2 Reproduction Commands

```
cd CSIT-8-PROJECT
python3 -m venv venv && source venv/bin/activate
```

```
pip install -r requirements.txt

python src/utils/preprocess_geolife.py --max-users 50

python src/experiments/run_experiments.py \
    --max-trajectories 10 \
    --compression-ratios 2.0 5.0 10.0 20.0 \
    --algorithms dp vw squish rw greedy_policy proposed \
    --data-file data/processed/trajectories.pkl

python src/experiments/generate_dataset_plots.py \
    --data-file data/processed/trajectories.pkl

python src/experiments/generate_plots.py \
    --results-file results/experiment_results.csv \
    --trajectories-file data/processed/trajectories.pkl
```

A.3 Verification

After experiments, `results/experiment_results.csv` should contain 240 rows for the command above; proposed rows should have non-null `turn_preservation` and `stop_preservation`. Key figures: `trajectory_comparison.png`, `compression_error_curves.png`, `metric_comparison_5x.png`, `dataset_turns_stops.png`.

End of Report